Luke Liang

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Education

Columbia University, School of Engineering and Applied Science — New York, NY
M.S. in Computer Science, Machine Learning Track, GPA 3.66 — *Expected Dec 2026*

Miami University, College of Engineering and Computing — Oxford, OH
B.S. in Computer Science, B.S. in Data Science & Statistics, Minor in Math, GPA 3.86 — *Aug 2021 - May 2024*

- Astronaut Scholarship (national), Summa Cum Laude, Honors College, ML training on Oak Ridge National HPC

Experience

Columbia University — Sept 2025 – Present
Research Assistant — *New York, NY*

- Automated a bioinformatics pipeline for high-throughput viral genomic sequencing, successfully processing 400+ patient samples and 5+ TB of data
- Transformed complex genomic outputs into actionable clinical insights, developing automated visualization scripts to generate diagnostic charts and variant summary dashboards

VISA — May 2026 – Aug 2026
Software Engineering Intern — *Austin, TX*

- Architected an AI-driven Container Vulnerability Remediation Agent leveraging Claude via MCP, cutting manual time-to-patch by 40% across 1,500+ supported container images
- Automated end-to-end pipelines by provisioning backend database, API integration with production servers, GitHub integrations to parse repository manifest files, submit patch PRs, and triggers for SMTP status alerts
- Deployed a conversational AI assistant in MS Teams using Copilot Studio, enabling developers to query container vulnerability states and trigger automated remediation workflows directly from chat

Miami University — Aug 2021 – Oct 2024
Research Assistant — *Oxford, OH*

- Engineered a high-performance simulation platform for CDC researchers to evaluate youth suicide prevention policies, translated real-world public health phenomena into scalable quantitative models in close collaboration with domain experts, and built custom visualization tools for multi-variable scenario.
- Pioneered anomaly detection pipelines achieving up to 95% accuracy in identifying erroneous network simulations by developing novel graph-to-image encodings, few-shot learning methods, and compression techniques.
- Built a stochastic HIV viral simulation in Python and NumPy, integrating Stanford's Drug Resistance Database and clinical mutation statistics to model cell-to-cell infection dynamics under 1-, 2-, and 3-drug treatment regimens.

JPMorgan Chase & Co. — Jun 2022 – Aug 2022
Software Engineering Intern — *Chicago, IL*

- Migrated legacy AngularJS applications to React and managed infrastructure deployments, enhancing front-end performance, component maintainability, and delivery reliability across internal systems.
- Won 1st Place at the JPMorgan Code for Good Hackathon (2022), collaborating with a team of interns to design and deploy a full-stack web application for a non-profit partner.

Publications

- **L. Liang**, et al. An Artificial Intelligence Approach to Support Adolescent Suicide Prevention Initiatives in the United States. Artificial Intelligence in Medicine (2026)
- **L. Liang**, H. Phan, P.J. Giabbanelli. Experimental Evaluation of a Machine Learning Approach to Improve the Reproducibility of Network Simulations. Simulation (2024)
- Co-authored 3 additional papers on network simulation error detection and suicide prevention modeling

Skills

Languages: Python, Java, C/C++, R, SAS, SQL, JavaScript/TypeScript, HTML/CSS
AI/ML & Data: PyTorch, scikit-learn, pandas, NumPy, Tensorflow, MATLAB
AI, Web, Cloud & Databases: React, Node.js, mySQL, Supabase, AWS Bedrock/Lambda/S3, Vercel
Developer & AI Tools: Claude Code, OpenAI Codex, Git/GitHub, Linux/Unix